

ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE

School of Computer and Communication Sciences

Handout 22

Project (Theory)

Principles of Digital Communications

Apr. 29, 2026

PROBLEM 1. Suppose $U = [U_1, \dots, U_n] \in \mathbb{R}^n$ is random vector uniformly distributed on the unit sphere. [Note: the unit sphere is $\{x \in \mathbb{R}^n : \|x\|^2 = 1\}$; this is the surface of the unit ball $\{x \in \mathbb{R}^n : \|x\|^2 \leq 1\}$.] A convenient way to obtain such a U is to let $X = [X_1, \dots, X_n] \sim \mathcal{N}(0, I_n)$, and then form $U = X/\|X\|$. Clearly U thus obtained will be on the unit sphere; to see that U is indeed uniformly distributed on this sphere observe that the probability density function $f_X(x) = (2\pi)^{-n/2} \exp(-\|x\|^2/2)$ of X is a function only of the norm $\|x\|$, and so X has no directional preference.

- (a) For $-1 < t < 1$, show that

$$\Pr(U_1 \geq t) = E\left[Q\left(\frac{t}{\sqrt{1-t^2}} \sqrt{X_2^2 + \dots + X_n^2}\right)\right].$$

Hint: consider $\Pr(U_1 \geq t \mid [X_2, \dots, X_n] = [x_2, \dots, x_n])$.

- (b) for $0 \leq t < 1$, show that

$$\Pr(U_1 \geq t) \leq \frac{1}{2}(1 - t^2)^{(n-1)/2}.$$

Hint:

(i) For $z \geq 0$, $Q(z) \leq \frac{1}{2} \exp(-z^2/2)$.

(ii) If $Z \sim \mathcal{N}(0, 1)$, then $E[\exp(-\lambda Z^2/2)] = 1/\sqrt{1 + \lambda}$.

- (c) Suppose $v \in \mathbb{R}^n$ is a unit vector. Show that $\langle U, v \rangle$ has the same distribution as U_1 .

PROBLEM 2. Suppose $c_1, \dots, c_m$ are vectors in $\mathbb{R}^n$ with $\|c_1\|^2 = \dots = \|c_m\|^2 = \mathcal{E}$, and they are used as the codewords of a communication system for an AWGN channel. Assume that the m messages are equally likely.

For a received vector $y \in \mathbb{R}^n$, let $\text{score}(i, y) = \frac{\langle y, c_i \rangle}{\|c_i\| \|y\|}$. By the Cauchy-Schwartz inequality, score takes values in the interval $[-1, +1]$.

- (a) Show that the receiver deciding $\hat{i} = \arg \max_i \text{score}(i, y)$ minimizes the probability of error.
- (b) Suppose message 1 is sent so that $Y = c_1 + Z$ is the received vector, where $Z \sim \mathcal{N}(0, \sigma^2 I_n)$. Let $-1 < t < 1$. Show that

$$\Pr(\text{score}(1, Y) \leq t) = \Pr\left(\sqrt{\mathcal{E}} + Z_1 \leq \frac{t}{\sqrt{1-t^2}} \sqrt{Z_2^2 + \dots + Z_n^2}\right)$$

- (c) For a given energy per bit $\mathcal{E}_b$, and a given bits/dimension ρ , let $m = \lceil 2^{\rho n} \rceil$, and $\mathcal{E} = \mathcal{E}_b \log_2 m \geq n\rho\mathcal{E}_b$. With Y still as in part (b), show that if ρ , $\mathcal{E}_b$, σ^2 and t satisfy $\rho\mathcal{E}_b/\sigma^2 > t^2/(1-t^2)$, then

$$\Pr(\text{score}(1, Y) \leq t) \rightarrow 0 \quad \text{as } n \text{ gets large.}$$

Hint: by (b), we know that $\Pr(\text{score}(1, Y) \leq t)$ is equal to the probability of the event

$$\sqrt{\frac{\mathcal{E}}{n}} + \frac{Z_1}{\sqrt{n}} \leq \frac{t}{\sqrt{1-t^2}} \sqrt{\frac{Z_2^2 + \dots + Z_n^2}{n}},$$

and by the law of large numbers we know that $\frac{Z_2^2 + \dots + Z_n^2}{n} \rightarrow \sigma^2$ as n gets large.

Suppose now the n -dimensional codewords $c_1 \in \mathbb{R}^n, \dots, c_m \in \mathbb{R}^n$ are chosen in a ‘random’ fashion. Namely, each c_i is chosen uniformly distributed on the n dimensional sphere of radius $\sqrt{\mathcal{E}}$, and $c_1, \dots, c_m$ are chosen independently. Since the codewords are now random vectors, it is notationally more appropriate to denote them as $C_1, \dots, C_m$ (as per our habit of using capital letters for random objects). Observe that each C_i can be obtained by first generating a U that is uniformly distributed on the unit sphere (as in the Problem 1 above) and then scaling it by $\sqrt{\mathcal{E}}$.

- (d) Suppose again that message 1 is sent so that $Y = C_1 + Z$. Note that Y is independent of $C_2, \dots, C_m$. Show that

$$\Pr\left(\max_{2 \leq i \leq m} \text{score}(i, Y) \geq t\right) \leq \frac{1}{2}(m-1)(1-t^2)^{(n-1)/2}$$

Hint: the event in question is the union of $m-1$ events. Use Problem 1(b,c) above to bound the probability of each of them.

- (e) As in part (c), let ρ denote the bits per dimension so that $m = \lceil 2^{\rho n} \rceil$. Show that if ρ and t satisfy $(1-t^2) < 2^{-2\rho}$, then

$$\Pr\left(\max_{2 \leq i \leq m} \text{score}(i, Y) \geq t\right) \rightarrow 0 \quad \text{as } n \text{ gets large.}$$

- (f) Observe that $(1-t^2) < 2^{-2\rho}$ is the same statement as $t^2/(1-t^2) > 2^{2\rho} - 1$. As in part (c), let ρ denote the bits/dimension and $\mathcal{E}_b$ the energy per bit. Show that if ρ and $\mathcal{E}_b$ satisfy $\rho\mathcal{E}_b/\sigma^2 > 2^{2\rho} - 1$, then for any desired error probability ϵ , for large enough n we can find a code with $m = \lceil 2^{\rho n} \rceil$ codewords $c_1, \dots, c_m$, each codeword $c_i \in \mathbb{R}^n$ satisfying $\|c_i\|^2 = n\rho\mathcal{E}_b$, and the code having average error probability at most ϵ .

Hint: if $\rho\mathcal{E}_b/\sigma^2 > 2^{2\rho} - 1$ then we can find a t such that $\rho\mathcal{E}_b/\sigma^2 > t^2/(1-t^2) > 2^{2\rho} - 1$. Let ‘rule-A’ be the following decoding rule: ‘given received vector y , check if there is an i_0 such that $\text{score}(i_0, y) \geq t$ and $\max_{i: i \neq i_0} \text{score}(i, y) < t$; if there is such an i_0 decide that i_0 was the sent message. Note that the optimal rule in (a) will have a smaller error probability than this rule. What do (c) and (d) say about the error probability of rule-A?’

- (g) (Optional numerical study.) Fix $\rho = 0.5$ and $\sigma^2 = 1$. For each $\mathcal{E}_b = 3, 4, 5, 6$ and for each $n = 20, 24, 28, 32$, generate random codes as above, and estimate the error probability of these codes by simulating their performance (e.g., by trying a code 500 times and checking how many times the rule in part (a) decides incorrectly).

PROBLEM 3. Consider the following four operations defined on two dimensional vectors:

$$T_1 : x \rightarrow x, \quad T_2 : (x_1, x_2) \rightarrow (-x_2, x_1), \quad T_3 : x \rightarrow -x, \quad T_4 : x \rightarrow -T_2(x).$$

(If we think of $x = (x_1, x_2)$ as the complex number $x_1 + jx_2$, the operations are multiplications by 1, j , -1 and $-j$.) For an even positive integer n , channel input $x \in \mathbb{R}^n$, and one of the above four operations T , we define $T(x)$ to mean $[T(x_1, x_2), T(x_3, x_4), \dots, T(x_{n-1}, x_n)]$, e.g., $T_2(1, 2, 3, 4) = (-2, 1, -4, 3)$.

For the rest of the problem suppose n is a positive even integer. Suppose we have a channel where the output $Y \in \mathbb{R}^n$ is obtained from the input $x \in \mathbb{R}^n$ by first choosing T as one the transformations $T_1, \dots, T_4$ above (each equally likely) and forming

$$Y = T(x) + Z \quad \text{where } Z \sim \mathcal{N}(0, \sigma^2 I_n) \text{ is independent of } T.$$

- (a) Suppose we have a code with m codewords $c_1, \dots, c_m$ in $\mathbb{R}^n$ with the property that if c is a codeword, then $-c$ is also a codeword. Show that if we use this code with equally likely messages on the channel above, the error probability is at least $1/2$.
- (b) Consider a code constructed randomly as in Problem 2. Suppose this code is used for the channel above. Show that the error probability is upper bounded by at most 4 times the upper bound on its error probability derived in Problem 2 (where a channel was $Y = x + Z$ — without a T).

Hint: Define the score of a message i as $\max_{1 \leq k \leq 4} \langle y, T_k(c_i) \rangle / (\|y\| \cdot \|T_k(c_i)\|)$.

Hence, if $\rho \mathcal{E}_b / \sigma^2 > 2^{2\rho} - 1$, by taking n large enough, we can reliably transmit ρ bits/dimension and keep to an energy budget of $\mathcal{E}_b$ per bit, e.g., if $\rho = 0.5$, we need $\mathcal{E}_b > 2\sigma^2$.

Suppose we are given a code that has probability of error δ when it is used on the channel $Y = x + Z$. We want to use this code on the channel $Y = T(x) + Z$ as above. As we have have seen in (a) there is no guarantee that this will lead to a reliable system.

- (c) Suppose we have access to an oracle who tells the receiver which one of the T_i 's was chosen at T — but the oracle might be in error with probability δ_1 . How can we implement a system with error probability at most $\delta + \delta_1$?
- (d) Show that by first transmitting a short known sequence, e.g., A, A, A, A , then followed by a codeword, one can implement such an oracle.

Hint: estimate which T was chosen from the first few channel outputs.